

Copilot Studio Training – Day 11

Agenda:

- Multi Agent Capabilities in Copilot Studio
- Analytics in Copilot Studio
- Transcript Conversations in Copilot Studio
- Log Telemetry using Azure Application Insights
- M365 Copilot Introduction & Comparison

Copilot Studio – Multi-Agent Architecture

1. Introduction

- Copilot Studio now supports **multi-agent setups**.
- Agents can be:
 - **Child Agents** – lightweight, embedded in the parent agent.
 - **Connected Agents** – standalone, independently published agents.

2. Key Differences: Child Agent vs Connected Agent

Feature	Child Agent	Connected Agent
Deployment	Part of main agent	Separate published agent
Reusability	Limited to parent agent	Can be reused in multiple agents
Inherits Settings	Yes	No
Configuration Complexity	Simple	More configuration needed
Use Case	Break down complex logic	Integrate mature/independent agents
Environment Requirement	Same Power Platform environment only	Same environment for connection

3. Setup Process

A. Create Main Agent

- Example: “Enquiry and Order Agent”
- Add **portal knowledge base**
- Use **generative AI** to respond based on portal

B. Add Child Agent

- Go to Agents > Add Agent > *Create lightweight agent*
- Example: “Order Agent”
- Trigger: “Agent chooses based on description”
- Optional: Add tools (e.g., **Dataverse list rows** for order info)

1. Instruction Setup:

In the main agent, update instruction:

If user wants to know about orders, call /Order Agent.

Connect Existing Agent (Connected Agent)

- Go to Agents > Add Agent > *Connect Existing Agent*
- Options: Copilot Studio / Azure AI Foundry / M365 (Preview)
- Choose an already published agent
- Optionally pass conversation history

Instruction Setup:

Update main agent:

For credit-related queries, call connected agent.

- Enable: ☒ "Connect to and use this agent"

Important Notes

- Only agents in the **same environment** can be connected.
- Connected agents need to be **republished after changes**.
- You can **mix** child + connected agents depending on solution needs.

References

- Microsoft Docs on multi-agent capabilities
- Portal configuration and Dataverse integration guides

Use Case 01: TravelMate – AI Assistant for Dubai Tourists

Overview

TravelMate is a public-facing AI assistant that helps global tourists plan their trip to **Dubai**, with multi-agent orchestration using:

- A main agent for all travel inquiries
- A child agent for visa-related queries integrated with UAE visa portal

- Logging entries in **SharePoint** for every visa inquiry

Agent Architecture

Agent Name	Type	Functionality
TravelMate	Master Agent (Already created in Previous Class)	Smart travel assistant for all Dubai-related tourist services
Visa Agent	Child Agent	Handles visa inquiries using UAE portal & logs query to SharePoint

Agent Details

1. Master Agent: TravelMate

- **Description:**
A virtual assistant to help tourists plan their Dubai vacation, offering recommendations on hotels, attractions, transportation, events, etc.
- **Instructions Snippet:**
Already created in the Previous Class with Multilingual Capability.

2. Child Agent: Visa Agent

- **Trigger:** Agent decides based on description
- **Description:** Handles all visa-related queries and uses UAE visa public site
- **Knowledge Source:**
 - **UAE Visa Portal** → <https://www.uaevisa.ae>
- **Instructions:**

“When someone asks you a visa-related query, use the Knowledge and try to find out the answer from it. At the end, create an entry in SharePoint. Title is the name for the person who inquired.”

 **Instructions**

 

When someone asks you Visa related query use the Knowledge and try to find out the answer from it. At the end create an entry using  **Create item** . Title is the name for the person who inquired for.

- **Tool Integration:**
 - **SharePoint List:** Logs every inquiry for record-keeping and follow-up

1. Book a holiday

2. Help me for the Visa now. How many days of Visa do I need? and What's the pricign for that?

Reference:

<https://learn.microsoft.com/en-us/microsoft-copilot-studio/authoring-add-other-agents>

Use Case 02 – Multi-Agent Copilot (Connected Agent Use Case)

Based on the two previous use cases explained to your students, here's a new **Use Case 02** designed around **Connected Agents** — demonstrating how two independently functional agents (already built: **Price Mate** and **Auto PO Creator**) can now collaborate in a **multi-agent orchestration scenario**.

Title: Connected Intelligence – Price Quotation + Automated Purchase Order Generation

Powered by Connected Agents: Price Mate & Auto PO Creation Agent

Objective

Demonstrate how two separate business processes — *quotation* and *purchase order approval* — can operate seamlessly using **connected agents** within Copilot Studio. This enables a unified, modular architecture that improves reusability, separation of concern, and AI governance.

Business Scenario

A supplier sends a product inquiry email with multiple product codes. The AI system (Price Mate) responds with prices and availability. If the recipient confirms, the AI then hands off the request to another agent (Auto PO Creator), which generates the purchase order, performs currency conversion, and sends approval emails accordingly.

🔗 Architecture Overview

Agent Name	Type	Purpose
Quotation Manager	Main Agent	Receives and understands user intent, routes to proper agents
Price Mate	Connected Agent	Parses product inquiry, fetches pricing, returns summary
Auto PO Creator	Connected Agent	Creates PO in SharePoint, triggers approval, sends summary

Main Agent: Quotation Manager

Instructions:

If a user asks for product prices, call /Price Mate.

If a user confirms and asks to raise a PO, call /Auto PO Creation Agent.

Do not repeat tasks. Route intelligently based on conversation history.

Test Conversation Simulation

User: "Please send me price for CAR-PART-0021 and CAR-PART-0040."

→ Quotation Manager calls → /Price Mate

→ Responds with HTML quotation including price, INR rate, and availability

User: "Great. Please raise the PO for these."

→ Quotation Manager calls → /Auto PO Creation Agent

→ PO is created, approval triggered (based on amount), and summary email sent

Agent Reusability Strategy

Agent	Reusable In
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Price Mate	Any quotation, tender, supplier interaction flow
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Auto PO Creator	Internal purchasing, vendor onboarding, finance
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Quotation Manager Could be replaced by another AI copilot or embedded chat

What are Analytics in Copilot Studio?

- Provides **visual reports** on bot performance and user behavior.
 - Helps **optimize bot responses** and **identify issues**.
 - Uses **Power BI dashboards** for insights.
 - Tracks **session engagement, resolution rates, escalations, and errors**.
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Demo Highlights

- **Accessing Analytics Tab** in Copilot Studio.
 - **Viewing bot performance through Power BI dashboards**.
 - **Filtering reports by date range**.
 - **Downloading conversation logs for session analysis**.
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Benefits of Copilot Studio Analytics

- ✓ **Improves bot efficiency** by tracking unresolved conversations.
- ✓ **Enhances user experience** by identifying **pain points**.

- ✓ **Optimizes workflows** based on **real user interactions**.
- ✓ **Helps refine GPT model responses** for better accuracy.

Reference: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/analytics-summary>

<https://azure.microsoft.com/en-us/pricing/calculator/?service=monitor>

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Conversation Transcripts in Copilot Studio

What is a Conversation Transcript?

- A **chat log** between a **user and an agent**.
- Stored in **Dataverse** in a table named **Conversation Transcript**.
- Captures chat history in **JSON format** with metadata like:
 - **Conversation start time**
 - **Bot GUID**
 - **User queries and bot responses**

Why is it Important?

- Normally, once a chat session is closed, **there's no way to retrieve it**.
- The **Conversation Transcript table** provides a way to **store and retrieve** chat history for analysis, troubleshooting, and compliance.
- Helps in **debugging issues** and understanding **bot interactions**.

How to Access Conversation Logs

1. **Navigate to Dataverse**
 - Go to **make.powerapps.com**.
 - Search for "**Conversation Transcript**" in **Tables**.
2. **Extracting Chat Logs**
 - The **Content column** stores chat details in **JSON format**.
 - Copy and format the JSON to view structured data (e.g., using Visual Studio Code).
3. **Understanding JSON Structure**
 - Contains **activities** like:
 - **Bot responses**
 - **User queries**.
 - **Session info** (start time, end time).
 - Metadata includes:
 - **Bot ID and Tenant ID**.
 - **Conversation flow details**.

Security & Access Control

- **Security Role: Bot Transcript Viewer**

- Provides **read-only access** to conversation logs.
- **Cannot be customized** (default security setting).
- Access is limited to **user-level** transcripts (users can only see their own data).
- **Sharing Copilot with Users**
 - When sharing a **Copilot**, users can be assigned **Bot Transcript Viewer** role.
 - This restricts access to viewing only their own conversation logs.

Use Cases & Benefits

- **Tracking user interactions** for analytics.
- **Troubleshooting bot performance** and errors.
- **Extracting conversation logs for compliance** (e.g., GDPR data requests).
- **Understanding bot engagement** across multiple deployed copilots.

Demo Highlights

- **Extracting chat logs** from Dataverse.
- **Formatting JSON** for readability.
- **Sorting transcripts by conversation start time.**
- **Assigning security roles** to users for controlled access.

Conversation transcripts are available for download about 36 minutes after an analytic session ends. You can download any time period within the last 29 days. You can download them in [Dataverse via the Power Apps portal](#) and as [session chat transcripts via the Copilot Studio app](#). It can take up to an hour after the analytic session ends before the related data appears on the analytics dashboard.

Application Insights with Copilot Studio

What is Application Insights?

- A **SaaS-based** service in Microsoft Azure for monitoring applications.
- Collects **telemetry data** on:
 - Web server performance
 - Web page interactions
 - Errors, warnings, and general app health

Why Use Application Insights?

- Tracks **user behavior** in web, mobile, and chatbot applications.
- Monitors **performance issues** and logs errors/warnings.
- Works with **Power Apps, Power Virtual Agents, and Copilot Studio**.

Integration Steps

1. **Obtain Connection String**
 - Go to **portal.azure.com** → Search for **Application Insights**.
 - Create a new resource or use an existing one.
 - Copy the **Connection String**.
2. **Configure in Copilot Studio**
 - Open **Copilot Studio** → Go to **Settings** → **Advanced**.
 - Paste the **Connection String** in the Application Insights section.
 - Enable logging settings.
3. **Publish and Monitor**
 - **Save & Publish** the bot after integrating.
 - View telemetry logs in **Application Insights** → **Usage & Events**.

Logging Options in Copilot Studio

1. **Log Activities:** Tracks all incoming and outgoing messages.
2. **Log Sensitive Activity Properties:** Includes **user ID, name, and text** (can be disabled for privacy).
3. **Log Node Actions:** Captures actions happening within a **topic node**.

Custom Telemetry Logging

- Allows tracking **specific events** beyond standard logging.
- Example: **Custom Telemetry Events** like “Power Tech Spec Event” to capture user interactions.

Using Application Insights Data

- View **User Activity Reports** (number of users, regions, sessions).
 - Analyze **event logs** (e.g., bot response time, errors).
 - **Troubleshoot issues** using:
 - **Conversation Timeline:** Shows the flow of user interactions.
 - **Error Logging:** Captures loop errors or conversation failures.
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Demo Highlights

- **Integration of Application Insights with Copilot Studio.**
- **Real-time tracking** of chatbot interactions.
- **Error simulation:** Creating a circular loop to force an error and track it in logs.
- **Custom event logging** for deeper insights into chatbot behavior.

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